

CY 344: Food Chemistry: Processing, Preservation and Storage

Course Code	CY 344
Course Name	Food Chemistry: Processing, Preservation and Storage
L-T-P-C	3-0-0-3
Prerequisites	Consent of the faculty member
Intended for	B. Tech.
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Water** Water microstructure, Availability in Foods, Water activity, Food Stability, Moisture sorption, State diagrams and freezing, Molecular mobility and the glass transition, Example: Candy manufacturing, Emulsions and foams.
- **Proteins** Amino Acids / Basic Building Blocks, Peptides and Proteins, Protein denaturation, Protein functionality, Emulsification, Nutritional Properties, Protein Modification, Processing and Storage, Browning reactions in foods.
- **Carbohydrates** Monosaccharides, Disaccharides and Oligosaccharides, Polysaccharides, related reactions, dietary carbohydrates and their sources, functional properties of dietary carbohydrates.
- **Lipids** Fatty acids, Glycerides, fat and oil processing, fatty acids in foods, rancidity and reversion of oils, emulsifying fats.
- **Enzymes** Specificity, Catalysis and Regulation, Temperature and pH, Water Activity, Electrolytes, endogenous enzymes in food, enzymes as food processing aids.
- **Vitamins and Minerals** Fat and water soluble vitamins, vitamins as food ingredients, major and trace minerals, minerals and canned food.
- **Flavor** Taste, odor, description of food flavors.
- **Texture** Its importance in food industries.
- **Color, additives and contaminants** A brief introduction
- **Food spoilage** Causes and remedies
- **Various techniques for food processing and preservations**
A brief introduction on food laws.

Text and Reference Books

1. John M. Deman, **Principles of Food Chemistry**.
2. Srinivasan Damodaran, Kirk Parkin, Owen R. Fennema, **Fennistrema'S Food Chemistry**, 4th Edition
3. David E. Newton, **Food Chemistry**
4. Tom P. Coultate, **Food: The Chemistry Of Its Components**
5. Meenakshi Paul, **Experimental Food Chemistry**.
6. Shalini Saxena, **Food Chemistry**.
7. Meenakshi Paul, **Effects Of Food Processing On Bioactive Compounds**
8. B. Sivasankar, **Food processing and preservation**.
9. R. Chaudhary, **Basics Of Food Chemistry**.
10. Frank A. Lee, **Basic Food Chemistry**
11. D. W.S.Wang, **Mechanism and theory in food chemistry**
12. J. Kaur, **Fundamentals of food chemistry**